

# Chapter 26

## Foil Factor Monomials/Polynomials (Molly-Polly)

|   |   |   |   |   |   |   |   |   |    |
|---|---|---|---|---|---|---|---|---|----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|----|

1)  $(x - 4y)^2 - x^2 - 4y^2 =$

- A) 0                      B)  $12y^2$                       C)  $-2x^2$                       D)  $-2x^2 - 12y^2$                       E)  $12y^2 - 8xy$

2) If  $x^2 - y^2 = 284$ , and  $x - y = 2$ , **what is the value of  $x + y$ ?**

- A) 2                      B) 12                      C) 92                      D) 142                      E) 8

3) If  $(x - y)^2 = 1$  and  $x^2 + y^2 = 19,801$ , **what is the product of  $x$  and  $y$ ?**

- A) 1                      B) 12                      C) 9900                      D) 18                      E) 384,901

4)  $xy^2 + 3x^2y + 3(-xy^2 - x^2y^2) + 2xy^2 =$

- A)  $3(x^2y - x^2y^2)$   
B)  $2(x^2y - x^2y^2)$   
C)  $5x^2y^2$   
D)  $5(xy^2 - x^2y^2)$   
E)  $3xy^2 - x^2y^2$

5) If  $x^2 - y^2 = 64$  and  $x - y = 2$ , **what is the value of  $x$ ?**

- A) 8                      B) 10                      C) 11                      D) 17                      E) 18

|   |   |   |   |   |   |   |   |   |    |
|---|---|---|---|---|---|---|---|---|----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|----|

**6)** Assuming  $v$  is a **positive integer**, what is the sum of all possible integer roots to the equation;  $x^2 - vx + 30$ ?

- A) 41                      B) 11                      C) 59                      D) 30                      E) 72

**7)** If  $x^2 - y^2 = x + y = 39$ , **what is the value of  $x - y$** ?

- A) 0                      B) 25                      C) 17                      D) 13                      E) 1

**8)** If  $g(x) = (x)^2 - (j)^2 = 0$  and  $x \neq j$ , which of the following **must be the sum of  $x$  and  $j$** ?

- I.  $2x$   
 II.  $2j$   
 III.  $0$

- A) I only  
 B) II only  
 C) III only  
 D) I and II  
 E) All of the above

**9)**  $(2x^2)^4(x^3)$  **is equal to** which of the following?

- A)  $2x^{24}$                       B)  $4x^{24}$                       C)  $8x^{11}$                       D)  $16x^{11}$                       E)  $8x^{24}$

**10)** Which of the following is the **simplified form** for the following polynomial equation;

$$(4c^2 + 3c - 2) - (12c^2 - 3 + 4c)?$$

- A)  $8c^2 + 6c + 1$   
 B)  $-48c^2 + 6c + 1$   
 C)  $-8c^2 - c + 1$   
 D)  $-8c^2 - c + 5$   
 E)  $-8c^2 + 6c - 1$

# Solutions Chapter 26

## Foil Factor Monomials Polynomials

### Molly Polly

- |      |       |
|------|-------|
| 1) E | 6) E  |
| 2) D | 7) E  |
| 3) C | 8) C  |
| 4) A | 9) D  |
| 5) D | 10) C |

### Molly Polly – made up name for this chapter

1) To simplify, **foil the quantity squared**.  $(x - 4y)^2 = (x - 4y)(x - 4y) = x^2 - 4xy - 4xy + 16y^2 = x^2 - 8xy + 16y^2$ . Combine with the rest of the question;  $x^2 - 8xy + 16y^2 - x^2 - 4y^2 = 12y^2 - 8xy$

12y<sup>2</sup> - 8xy **Answer E** Difficulty Level 5

2) If a question contains an expression that includes the difference of squares, it is usually a good plan to factor the expression first. Since  $x^2 - y^2$  can be factored as the difference of squares into  $(x + y)(x - y)$ , substitute this into the given equation;  $x^2 - y^2 = 284$ ,  $(x + y)(x - y) = 284$ . Since the expression  $x - y = 2$  (given), solve for the expression  $(x + y)$ .

$$\begin{aligned} (?) \quad (2) &= 284 \\ (x + y)(x - y) &= 284 \end{aligned}$$

where  $x + y$  is the unknown. Using **(?)(2) = 284; divide by 2; ? = 142. Since ? = x + y, x + y = 142.**

142 **Answer D** Difficulty Level 8

3) When identifying the quantity of squares on the ACT, foiling is usually the best approach;  $(x - y)^2 = (x - y)(x - y) = x^2 - xy - xy + y^2 = x^2 - 2xy + y^2$ . Next, use the Commutative property of addition and substitute  $x^2 + y^2 = 19801$  into the given quantity;  $x^2 + y^2 - 2xy = 1$ ;  $19801 - 2xy = 1$ . Isolate  $xy$  because that is the product of  $x$  and  $y$ . Remember when variables or numbers are next to each other, they are to be multiplied together. After substituting, subtract 19801 from both sides of the equation;  $-2xy = -19800$ , then divide both sides by  $-2$ ;  $xy = 9900$

9900 Answer C Difficulty Level 8.5

4) Simplify first by distributing,  $xy^2 + 3x^2y + 3(-xy^2 - x^2y^2) + 2xy^2 = xy^2 + 3x^2y - 3xy^2 - 3x^2y^2 + 2xy^2$ , then identify and combine like terms. All the  $xy^2$ 's add to zero ( $1 - 3 + 2 = 0$ ). Therefore,  $3x^2y - 3x^2y^2$  remains. In a different form, factoring a 3 out of  $-3x^2y - 3x^2y^2 =$

$3(x^2y - x^2y^2)$  Answer A Difficulty Level 4

5) If a question contains an expression that includes the difference of squares, it is usually a good plan to factor the expression first. Since  $x^2 - y^2$  can be factored as the difference of squares into  $(x + y)(x - y)$ , substitute into the original equation;  $(x + y)(x - y) = 64$ . Since  $x - y = 2$  (Given)

$$\begin{aligned} (?) (2) &= 64 \\ (x + y)(x - y) &= 64 \end{aligned}$$

where  $x + y$  is the unknown. Using  $(?)(2) = 64$ ; divide by 2;  $? = 32$ . Since  $? = x + y$ ;  $x + y = 32$ . To find the value of  $x$ , use elimination or substitution. Using elimination;  $x + y = 32$

$$\begin{array}{r} x - y = 2 \\ \hline x + y = 32 \\ \hline 2x = 34 \\ x = 17 \end{array}$$

17 Answer D Difficulty Level 8

6) The given equation is a quadratic in standard form ( $Ax^2 + bx + c$ ). To find the roots of a quadratic, set it equal to zero, factor it, then solve it. The solution pairs of 30 are (1x30), (2x15), (3x10), and (5x6). The factored quadratics would be;

$$(x - 1)(x - 30) = x^2 - 31 + 30 \quad (1 \text{ and } 30)$$

$$(x - 2)(x - 15) = x^2 - 17 + 30 \quad (2 \text{ and } 15)$$

$$(x - 3)(x - 10) = x^2 - 13 + 30 \quad (3 \text{ and } 10)$$

$$(x - 5)(x - 6) = x^2 - 11 + 30 \quad (5 \text{ and } 6)$$

|         |               |       |
|---------|---------------|-------|
| Factors | Quadratic Eq. | Roots |
|---------|---------------|-------|

Notice all the factors are of the  $(x - a)$  variety. This ensures each equation has a **negative sign** in front of the middle term matching the equation given in this question;  $x^2 - vx + 30$ .

*For example*,  $(x + 2)(x + 15)$  could not be used because after foiling  $x^2 + 17x + 30$  has the middle term  $17x$ , which is positive. The equation given in the question is  $-vx$  therefore, the middle term of the quadratic must sum to a negative number.

Finally, add all the possible integer roots;

$$1 + 30 + 2 + 15 + 3 + 10 + 5 + 6 = 72$$

**72 Answer E Level 10**

7) A good plan of attack when seeing the difference of squares on the ACT is to factor. In this question,  $x^2 - y^2 = (x + y)(x - y)$ .

Since  $x^2 - y^2 = x + y = 39$ , then  $x - y = 1$

$$x^2 - y^2 = (x + y)(x - y)$$

$$(39) \times (1) = 39$$

**1 Answer E Difficulty Level 8**

8) Since any number squared is positive, the sign does not matter for  $x$  or  $j$ . The only possible values for  $x$  and  $j$  that satisfy both the equation;  $(x)^2 - (j)^2 = 0$  and the constraint  $x \neq j$  must be additive inverses(opposites) of each other. Remember, additive inverses always sum to zero. (Examples of additive inverses(opposites);  $-5$  and  $5$ ,  $2$  and  $-2$ ) To show this, plug in the first example;  $5$  and  $-5$ ; if  $x = 5$  and  $j = -5$ ;  
 $(5)^2 - (-5)^2 = 25 - 25 = 0$

III Only Answer C Difficulty Level 8

9) Remember when multiplying with the same base, add the exponents, and when taking a power to a power, multiply the powers together.  
 $(2x^2)^4 = 16x^8$ ;  $16x^8(x^3) = 16x^{11}$ . Do not forget that  $2$  must also be raised to the  $4^{\text{th}}$  power because it is inside the parenthesis.

$16x^{11}$  Answer D Difficulty Level 5

10) Since there is a negative sign between the two sets of parentheses, this question should be treated as subtraction. First, distribute the negative;  
 $(4c^2 + 3c - 2) - (12c^2 - 3 + 4c) = 4c^2 + 3c - 2 - 12c^2 + 3 - 4c$ ;  
 then combine like terms;

$-8c^2 - c + 1$  Answer C Difficulty Level 2.5